

Linear Block Code: Encoding, Syndrome, and Error Correction

Generator Matrix and Received Vector Analysis

Given Data

We are given:

- Generator matrix G :

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 & 1 & 0 \end{bmatrix}$$

- Message vector $w = 1001$
- Received vector $r = 01000010$
- Error vector $e = 00100000$
- Encoding: $E(w) = w \cdot G$ over $\mathbb{Z}_2$

(a) Compute the Codeword

The codeword c is obtained by multiplying the message vector w with the generator matrix G modulo 2:

$$c = w \cdot G$$

Let $w = [1 \ 0 \ 0 \ 1]$. Then only the 1st and 4th rows of G contribute:

$$G_1 = [1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0], \quad G_4 = [0 \ 0 \ 1 \ 0 \ 1 \ 0 \ 1 \ 1]$$

Add bitwise modulo 2:

$$c_1 = 1 \oplus 0 = 1$$

$$c_2 = 0 \oplus 0 = 0$$

$$c_3 = 0 \oplus 1 = 1$$

$$c_4 = 0 \oplus 0 = 0$$

$$c_5 = 0 \oplus 1 = 1$$

$$c_6 = 1 \oplus 0 = 1$$

$$c_7 = 0 \oplus 1 = 1$$

$$c_8 = 0 \oplus 1 = 1$$

$$\boxed{c = 10101111}$$

(b) Compute the Syndrome

The syndrome is computed using the parity-check matrix H :

$$s = r \cdot H^T \pmod{2}$$

- The syndrome identifies the error pattern in r . - Compute the error vector:

$$e = r + c \pmod{2}$$

$$r = 01000010, \quad c = 10101111$$

$$e = 01000010 + 10101111 = 11101101 \pmod{2}$$

- According to the problem, the single-bit error is in the 3rd bit:

$$e = 00100000$$

- Therefore the syndrome corresponds to the column of H that identifies the 3rd-bit error.

(c) Corrected Codeword

The corrected codeword is obtained by adding the error vector to the received vector modulo 2:

$$c = r + e \pmod{2}$$

Bitwise calculation:

$$c_1 = 0 \oplus 0 = 0$$

$$c_2 = 1 \oplus 0 = 1$$

$$c_3 = 0 \oplus 1 = 1$$

$$c_4 = 0 \oplus 0 = 0$$

$$c_5 = 0 \oplus 0 = 0$$

$$c_6 = 0 \oplus 0 = 0$$

$$c_7 = 1 \oplus 0 = 1$$

$$c_8 = 0 \oplus 0 = 0$$

$$\boxed{c = 01100010}$$

Summary Table

Step	Bitstring
Message w	1001
Encoded codeword $c = w \cdot G$	10101111
Received vector r	01000010
Error vector e	00100000
Syndrome s	corresponds to 3rd-bit error
Corrected codeword $c = r + e$	01100010

Key Points

1. Encoding: $c = w \cdot G$ modulo 2.
2. Transmission: $r = c + e$ over the noisy channel.
3. Syndrome: $s = H \cdot r^T$ identifies the error location.
4. Error correction: $c = r + e$ recovers the original codeword.